Electric Machines

Synchronous Motors

- **Controlling I_a**
  - if E_a is smaller than V_t, drawn current is lagging - therefore **absorbing** reactive power (+Q)
  - if E_a is larger than V_t, drawn current is leading **(supplying Q = -Q)**
    -typically operates by supplying Q instead of drawing Q
  - companies care about Q since oscilalting energy results in extra current flowing through transmission lines - use synchronous motors to reduce this

- **Synchronous Motor Ratings**
  - speed of rotation for a given frequency is determined by number of poles of the motor
  - Rating of synchronous motor given by VA and V

Example:

> You are at a country where the power supply has a frequency of 50 Hz but you are required to supply a load of 300 kW at 60Hz. You have with you synchronous motors and synchronous generators with all possible numbers of poles. If you decide to generate the 60Hz power with a set of a synchronous motor driving a synchronous generator. What is the number of poles that you need to have in the motor and in the generator in order to convert the available 50Hz Power to 60Hz Power?

$$n = \frac{120 \cdot f_e}{P}$$

$$n_m = \frac{120 \cdot (50)}{P_{poles_motor}} = \frac{6000}{P_{poles_motor}} \qquad n_g = \frac{120 \cdot (60)}{P_{poles_generator}} = \frac{7200}{P_{poles_generator}}$$

Since both machines move at the same speed then:

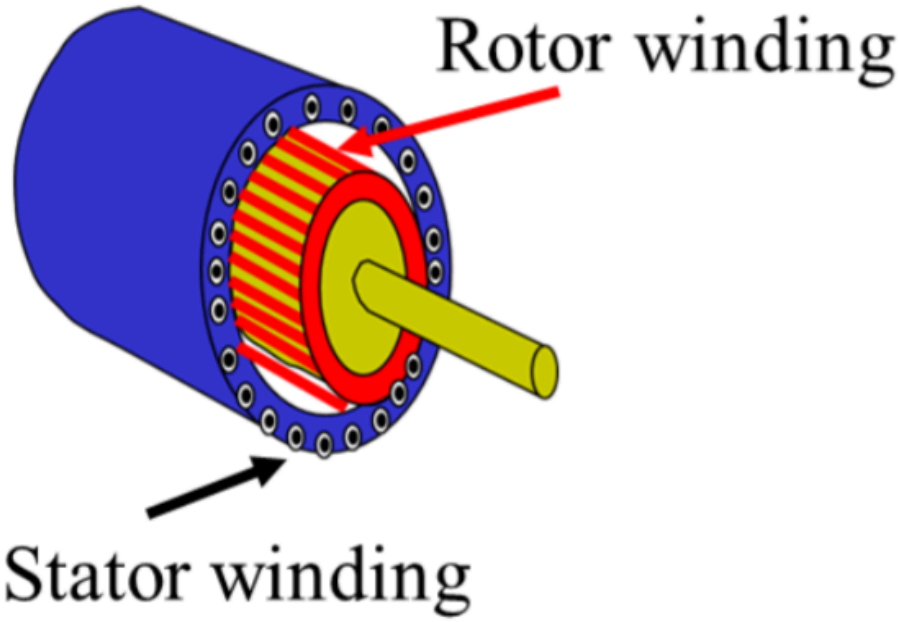
$$n_m = n_g \Rightarrow \frac{6000}{P_{motor}} = \frac{7200}{P_{generator}} \Rightarrow \frac{P_{generator}}{P_{motor}} = \frac{72}{60} = \frac{12}{10}$$

A 10 pole motor should be used with a 12 pole generator

Synchronous Generators

- **Frequency Regulation**
  -Load fluctuation causes frequency variations
  - High load results in lower frequency
  - Low load results in higher frequency

Induction Motor



- Two types of induction Machines
  - Wound rotor induction Machines
    - uses windings
  - Squirrel Cage
    - uses bars
    - easier and cheaper to build
    - more robust
    - not brushed so no sparks
- Operation of stator windings in induction motor is the same as synchronous motor
- Inducted voltage in the rotor is given by $e_{\text{induced}} = (V \times B) \cdot l$
  - V = velocity of stator winding flux
  - l = length of linked portion of rotor winding
  - B = flux density in rotor
- % difference between mechanical and electrical speed is known as **rotor slip** e.g. $s = \frac{n_s - n_{\text{mech}}}{n_s}$ where n_s is synchronous speed.

- Example:

The rotor in a 4-pole induction machine turns at 1400 rpm. What is the slip of the machine and the frequency of currents in the rotor?

$$n_s = \frac{120f}{p} \qquad s = 1 - \frac{n_{mech}}{n_s}$$

# Poles	Speed
2	3600
4	1800
6	1200
8	900

For a 4 pole machine, assuming 60 Hz operation, we know that the synchronous speed is 1800 Hz

Then:

$$s = 1 - \frac{1400}{1800} = 0.22$$

$$f_{e_rotor} = 0.22 \cdot f_{e_stator} = 13.2\ Hz$$

Differences

- Synchronous motor runs at constant speed
- Induction motor speed varies based on load